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Dossier "AIDS"

Humoral response in rabbits immunized with calcium phosphate adjuvanted HIV-1 gp160 antigen

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Summary – Rabbits were immunized with either calcium phosphate adjuvanted purified HIV-1 gp160 or a fluid preparation. Circulating antibodies were detected by ELISA, RIPA and Western Blot tests. Sera of rabbits immunized with the adjuvanted preparation contained high levels of anti-gp160 antibodies, as well as antibodies recognizing p55, p25 and p18. On the contrary, rabbits immunized with the fluid preparation contained only anti-p18 antibodies. Neutralizing antibodies were also detected. It is concluded that the calcium phosphate adjuvant could be used for preparation of candidate anti-HIV vaccines, since it permits one to induce high levels of circulating antibodies, in the absence of untoward reactions as observed when aluminium adjuvants or water in oil emulsions are used.

immunization / HIV-1 / gp160 / circulating antibodies / untoward reactions / calcium phosphate / aluminium / adjuvants

Introduction

Simple and combined DTP (Diphteria-Tetanus-Pertussis) and inactivated Polio vaccines and allergens absorbed into calcium phosphate have been used for many years for the immunization of children and adults or hyposensitization of allergic patients [19, 29, 30, 32, 36]. Field trials and animal experiments have been carried out using either simple or combined vaccines or simultaneous immunizations such as with BCG, yellow fever, measles and hepatitis B vaccines [4, 6, 7, 12, 15, 31, 34, 41]. Several programs have been studied in developing countries with the aim of simplifying vaccination schedules. The efficiency of a two-dose DT vaccination was ascertained, even in pregnant women [1, 33, 37, 41].

The use of adsorbed calcium phosphate preparations was found to be advantageous in various systems. The adverse local and generalized reactions were rare and mild especially by the choice of detoxified purified toxins, both in normal and allergic children and adults. Hyposensitisation carried out with house dust, mite and grass pollen adjuvanted extracts gave a high rate of satisfactory therapeutic results with a lower number of injections compared to fluid preparations. Total and specific IgE levels before and after immunotherapy were not significantly different [19].

On the contrary, data has been provided that aluminium compounds (hydroxide or phosphate) used as adjuvants had many drawbacks and increased levels of homocytotropic antibodies in animal species and IgE antibodies in man [9, 22, 23, 25, 42, 45]. An increased incidence of allergic disease following the periodic use of aluminium adsorbed vaccines has even been suggested [42]. Results on the use of calcium phosphate adsorbed vaccines and allergens prompted us to also investigate the immune response of the adsorbed HIV-1 gp160 antigen. The results of preliminary studies in rabbits are reported here.

Material and methods

Purified HIV-1 gp160 recombinant antigen was prepared by Transgène and PMSV (France). It was kindly donated by Prof D Zagury and Dr M Kaczorek.

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Calcium phosphate adjuvant was prepared according to a previously described method [30]. It can be purchased three times concentrated from Superfos Biosector a/s (Denmark).

Male white New Zealand rabbits of 3.5 kg weight were obtained from Iffa-Credo (France).

Elisa (ELAVIA-1) and Western Blot (NEW LAV BLOT-I) kits (Sanofi Diagnostic Pasteur, Marnes-la-Coquette, France) were used for detecting anti-HIV-1 rabbit antibodies with peroxydase labelled purified goat anti-human IgG antibody for human control sera and purified goat anti-rabbit IgG for rabbit sera. O-phenylenediamine dihydrochloride (OPD) and diaminobenzidine tetrahydrochloride (DAB) from Sigma were used as chromogenic agents in Elisa and Western Blot tests. Elisa results were read with the aid of a microplate reader with 492 and 620 nm filters (LP-200 Sanofi Diagnostic Pasteur) and absorbance (A) measured. Elisa for titration of rabbit anti-HIV-1 antibodies were also carried out on flat-bottomed 96-well polyvinyl chloride plates (Dynatech) coated with 100 µl of 5 µg/ml purified HIV-1 gp160 solution in 0.1 M carbonate buffer (pH 9.6) at 37°C for 2 hours. The same buffer solution was used for control wells. The plates were washed twice with the washing solution (R2 diluted 1/10) of the ELAVIA-1 kit. Titrations were carried out as with the ELAVIA-1 kit, using the same reference sera.

One thousand µg of gp160 were adsorbed at pH 6.8-7.0 onto 25 ml of the calcium phosphate adjuvant by mere mixing. The rabbits were immunized with 0.5 ml (20 µg) of the adjuvanted antigen.

The immunogen was given by the subcutaneous route into the flanks of the rabbits on day D0. Boosters were given as specified, *eg* on days 15 and 56, (D15 and D56), etc. Blood samples were taken from the ear vein on D0, D14, D28, etc., and sera collected after blood clotting.

Results

Four rabbits were immunized by two injections on D0 and D15; three rabbits, nos 2, 7 and 8 received the calcium phosphate adjuvanted (CaPh) gp160 preparation and one rabbit, no 3 the gp160 in saline.

Absorbance (A) of rabbit sera and the positive and negative controls of the ELISA kit were measured at a 1/100 dilution. Assays are valid if $A \geq 0.8$ for the positive control serum and $A < 0.2$ for the negative control serum. Serum samples are positive for HIV-1 antibodies if $A \geq 0.3$ and negative for $A < 0.3$.

Table I shows the results relative to A values for immunized rabbit sera collected until D84 and for the control human sera, analyzed together on the same day. The results for the positive and negative control sera of the kit were valid.

The absorbance of the serum for the rabbit immunized with the fluid preparation on D0 was slightly higher than that of the sera from rabbits given the CaPh adjuvanted preparation, probably because their serum was somewhat hemolytic. The A values of the sera from the four rabbits immunized on D0 and D14 increased until D28, but decreased for the fluid preparation until D56, when a booster was given (to two rabbits).

The HIV-1 gp160 coated ELISA plates yielded much higher absorbance values for the sera from rabbits immunized with the adjuvanted preparation than for the serum from the rabbits given the fluid preparation. Absorbance values were also

Table I. Results of ELISA: Absorbance (A) of rabbit anti HIV-1 gp160 sera.

Serums	*D0*	*D14*	*D28*	*D42*	*D56*	*D70*	*D84*	*ELISA with*
Rabbit no 2	0.050	0.021	0.956	1.462	3.298	3.102	2.533	
Rabbit no 3	0.232	0.575	1.519	0.825	0.739	2.810	2.795	ELAVIA-1
Rabbit no 7	0.017	0.122	1.660					
Rabbit no 8	0.010	0.285	1.354					A measured on the same day after the last bleeding
Human:								
Positive control				2.101				
Negative control				0.019				
Rabbit no 2	0.003	0.010	1.070	1.020	0.650	> 2.5	–	HIV-1 gp160 coated plates
Rabbit no 3	0.067	0.070	0.109	0.091	0.147	0.683	–	
Human:								A measured after each bleeding
Positive control	0.731	–	0.573	0.566	0.202	–	–	
Negative control	0.047	–	0.030	0.024	0.042	–	–	

lower for the HIV-1 gp160 coated plates, compared to those obtained with the ELAVIA-1 kit.

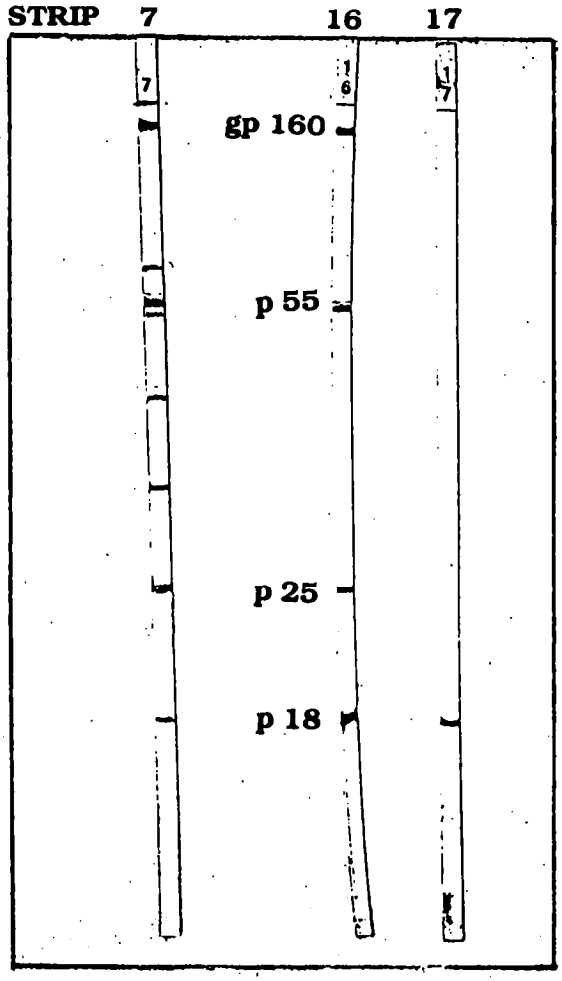

Fig 1. Reaction of anti-HIV-1 antibodies with the NEW LAV-BLOT-I kit. **Strip 7:** positive control serum, **Strip 16:** rabbit no 2 bleeding day 70 (Ca Ph adsorbed preparation), **Strip 17:** rabbit no 3 bleeding day 70 (fluid preparation).

Table II. Results of Western blots.

Bands with mobilities of:	*Sera of rabbits immunized with preparations*	
	CaPh adsorbed	*Fluid*
gp160-Glycoprotein	+	–
p-55 precursor of core proteins	+	–
p-25 core protein	+	–
p-18 core protein	+	+

Results of Western Blot assays are summarized in table II for all the tested sera and in figure 1 for the bleedings of rabbits no 2 (CaPh immunized) and no 3 (fluid preparation) and for the positive control serum. We see that the CaPh adsorbed preparation induced the formation of antibodies to the glycoprotein gp160 of the virus envelope, and to p55, p25 and p18 fractions. Only one p18 band was detected in the serum samples from the rabbit immunized with the fluid preparation.

In another experiment the rabbits were immunized with the adjuvanted HIV-1 gp160 preparation using the same schedule of injections, so as to verify the absence of adverse reactions. Western Blot analysis of the bleedings again revealed the presence of high levels of anti-gp160 HIV-1 antibodies.

The analysis of sera was also carried out by using radioimmunoprecipitation (RIPA) [46], high levels of antibodies were again detected. Neutralization of the LAI strain [2] was low or negative for a few bleedings, but low cross-neutralization was also found with a highly cytopathogenic strain from Zaïre (NDK).

Complementary analysis of rabbits no 2 serum, bleeding J87 and a mixture of bleedings at J114, J118, J121, J125 and J32 of four rabbits immunized at J0, J14, J29 and J107 gave again high RIPA antibody levels and complete virus neutralization for 1/50 and partial neutralisation for 1/400 serum dilutions [47].

Discussion and Conclusion

For the sera of rabbits immunized with the CaPh adjuvanted and the fluid purified HIV-1 gp160 antigen, about the same ELISA absorbance values (A) were found for blood samples taken on different days when analysed with a commercialized kit containing various HIV-1 antigens.

The A values were higher when sera were titrated on plates coated with the purified HIV-1 gp160 antigen for bleedings of the CaPh adsorbed preparation than for bleedings of the fluid preparation. However, values for both are lower than those measured with the kit containing the inactivated HIV-1 coated virus. These differences can be accounted for by assuming that ELISA results are not quantitative. Differences can arise from the varying amounts of antigens coated to the plates and the presence in the sera of antibodies

directed to gp-160 fractions in the preparations used for the immunizations.

Sera from rabbits immunized with CaPh adsorbed HIV-1 gp160 contained anti-gp160 antibodies but also against fractions having the mobility of p-55, p-25 and p-18 antibodies as detected by Western Blot. Sera from rabbits immunized with the fluid preparation contained only anti-p-18 antibodies, as found by Western Blot. The presence of high amounts of anti-gp160 antibodies were also detected with the radioimmunoprecipitation assay (RIPA), neutralizing antibodies were detected only at low levels or were even absent in some bleedings. This finding can tentatively be accounted for by assuming that the antibodies when present, are of low avidity as found for diphtheria and tetanus antitoxins early in the immunization schedule, and measured *in vivo* [26, 27]. These low avidity antibodies in animals vaccinated with adsorbed diphtheria vaccines were also recently found by titration on Vero cells [18].

It can be concluded that the calcium phosphate adjuvant should be used for potentiating candidate anti-HIV vaccines, since it permits to induce high levels of circulating antibodies, most of the time without the occurrence of untoward reactions [1, 4, 5, 15, 16, 28–30, 33, 35, 41] as are observed with aluminium adjuvants of water in oil emulsions [3, 8, 10, 11, 13, 14, 20, 24, 30, 38, 40, 43, 44].

No or mild reactions were encountered with calcium phosphate adjuvated vaccines in many field trials [29, 30, 32, 41] and mass vaccinations in the French Army [17], in Greece in the army and in the population [39] and by the Israel Ministry of Health in the Gaza Strip to eradicate sporadic cases of polio, with combined DTP inactivated Polio vaccine [21]. Calcium phosphate adjuvanted preparations have been commercially available in France for many years for the immunization of babies and adults and for hyposensitization treatments of allergic patients.

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